Atomic Bonding in Solids

2.5 BONDING FORCES AND ENERGIES

An understanding of many of the physical properties of materials is enhanced by a knowledge of the interatomic forces that bind the atoms together. Perhaps the principles of atomic bonding are best illustrated by considering how two isolated atoms interact as they are brought close together from an infinite separation. At large distances, interactions are negligible because the atoms are too far apart to have an influence on each other; however, at small separation distances, each atom exerts forces on the others. These forces are of two types, attractive (F_A) and repulsive (F_R), and the magnitude of each depends on the separation or interatomic distance (r); Figure 2.10*a* is a schematic plot of F_A and F_R versus r. The origin of an attractive force F_A depends on the particular type of bonding that exists between the two atoms, as discussed shortly. Repulsive forces arise from interactions between the negatively charged electron clouds for the two atoms and are important only at small values of r as the outer electron shells of the two atoms begin to overlap (Figure 2.10*a*).

The net force F_N between the two atoms is just the sum of both attractive and repulsive components; that is,

$$F_N = F_A + F_R \tag{2.3}$$

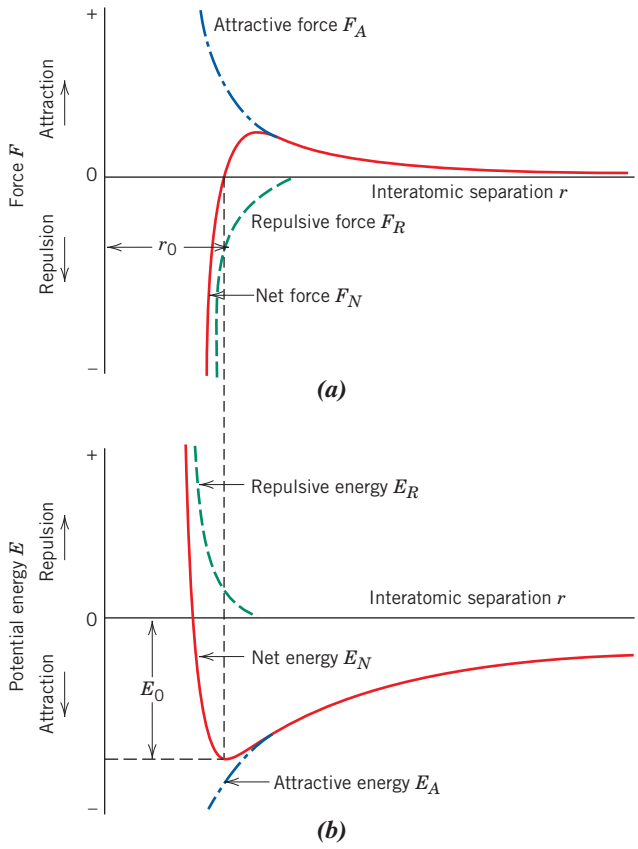


Figure 2.10 (*a*) The dependence of repulsive, attractive, and net forces on interatomic separation for two isolated atoms. (*b*) The dependence of repulsive, attractive, and net potential energies on interatomic separation for two isolated atoms.

which is also a function of the interatomic separation, as also plotted in Figure 2.10*a*. When F_A and F_R are equal in magnitude but opposite in sign, there is no net force—that is,

$$F_A + F_R = 0 \tag{2.4}$$

and a state of equilibrium exists. The centers of the two atoms remain separated by the equilibrium spacing r_0, as indicated in Figure 2.10*a*. For many atoms, r_0 is approximately 0.3 nm. Once in this position, any attempt to move the two atoms farther apart is counteracted by the attractive force, while pushing them closer together is resisted by the increasing repulsive force.

Sometimes it is more convenient to work with the potential energies between two atoms instead of forces. Mathematically, energy (E) and force (F) are related as

Force–potential energy relationship for two atoms

$$E = \int F\,dr \tag{2.5a}$$

And, for atomic systems,

$$E_N = \int_r^\infty F_N\,dr \tag{2.6}$$

$$= \int_r^\infty F_A\,dr + \int_r^\infty F_R\,dr \tag{2.7}$$

$$= E_A + E_R \tag{2.8a}$$

in which E_N, E_A, and E_R are, respectively, the net, attractive, and repulsive energies for two isolated and adjacent atoms.[5]

Figure 2.10*b* plots attractive, repulsive, and net potential energies as a function of interatomic separation for two atoms. From Equation 2.8a, the net curve is the sum of the attractive and repulsive curves. The minimum in the net energy curve corresponds to the equilibrium spacing, r_0. Furthermore, the **bonding energy** for these two atoms, E_0, corresponds to the energy at this minimum point (also shown in Figure 2.10*b*); it represents the energy required to separate these two atoms to an infinite separation.

bonding energy

Although the preceding treatment deals with an ideal situation involving only two atoms, a similar yet more complex condition exists for solid materials because force and energy interactions among atoms must be considered. Nevertheless, a bonding energy, analogous to E_0 above, may be associated with each atom. The magnitude of this bonding energy and the shape of the energy–versus–interatomic separation curve vary from material to material, and they both depend on the type of atomic bonding. Furthermore, a number of material properties depend on E_0, the curve shape, and bonding type.

[5]Force in Equation 2.5a may also be expressed as

$$F = \frac{dE}{dr} \tag{2.5b}$$

Likewise, the force equivalent of Equation 2.8a is as follows:

$$F_N = F_A + F_R \tag{2.3}$$

$$= \frac{dE_A}{dr} + \frac{dE_R}{dr} \tag{2.8b}$$

For example, materials having large bonding energies typically also have high melting temperatures; at room temperature, solid substances are formed for large bonding energies, whereas for small energies, the gaseous state is favored; liquids prevail when the energies are of intermediate magnitude. In addition, as discussed in Section 6.3, the mechanical stiffness (or modulus of elasticity) of a material is dependent on the shape of its force–versus–interatomic separation curve (Figure 6.7). The slope for a relatively stiff material at the $r = r_0$ position on the curve will be quite steep; slopes are shallower for more flexible materials. Furthermore, how much a material expands upon heating or contracts upon cooling (i.e., its linear coefficient of thermal expansion) is related to the shape of its E–versus–r curve (see Section 19.3). A deep and narrow "trough," which typically occurs for materials having large bonding energies, normally correlates with a low coefficient of thermal expansion and relatively small dimensional alterations for changes in temperature.

primary bond

Three different types of **primary** or chemical **bond** are found in solids—ionic, covalent, and metallic. For each type, the bonding necessarily involves the valence electrons; furthermore, the nature of the bond depends on the electron structures of the constituent atoms. In general, each of these three types of bonding arises from the tendency of the atoms to assume stable electron structures, like those of the inert gases, by completely filling the outermost electron shell.

Secondary or physical forces and energies are also found in many solid materials; they are weaker than the primary ones but nonetheless influence the physical properties of some materials. The sections that follow explain the several kinds of primary and secondary interatomic bonds.

2.6 PRIMARY INTERATOMIC BONDS

Ionic Bonding

ionic bonding

Ionic bonding is perhaps the easiest to describe and visualize. It is always found in compounds composed of both metallic and nonmetallic elements, elements situated at the horizontal extremities of the periodic table. Atoms of a metallic element easily give up their valence electrons to the nonmetallic atoms. In the process, all the atoms acquire stable or inert gas configurations (i.e., completely filled orbital shells) and, in addition, an electrical charge—that is, they become ions. Sodium chloride (NaCl) is the classic ionic material. A sodium atom can assume the electron structure of neon (and a net single positive charge with a reduction in size) by a transfer of its one valence 3*s* electron to a chlorine atom (Figure 2.11*a*). After such a transfer, the chlorine ion acquires a net negative charge, an electron configuration identical to that of argon; it is also larger than the chlorine atom. Ionic bonding is illustrated schematically in Figure 2.11*b*.

coulombic force

The attractive bonding forces are **coulombic**—that is, positive and negative ions, by virtue of their net electrical charge, attract one another. For two isolated ions, the attractive energy E_A is a function of the interatomic distance according to

Attractive energy—interatomic separation relationship

$$E_A = -\frac{A}{r} \tag{2.9}$$

Theoretically, the constant A is equal to

$$A = \frac{1}{4\pi\varepsilon_0}(|Z_1|e)(|Z_2|e) \tag{2.10}$$

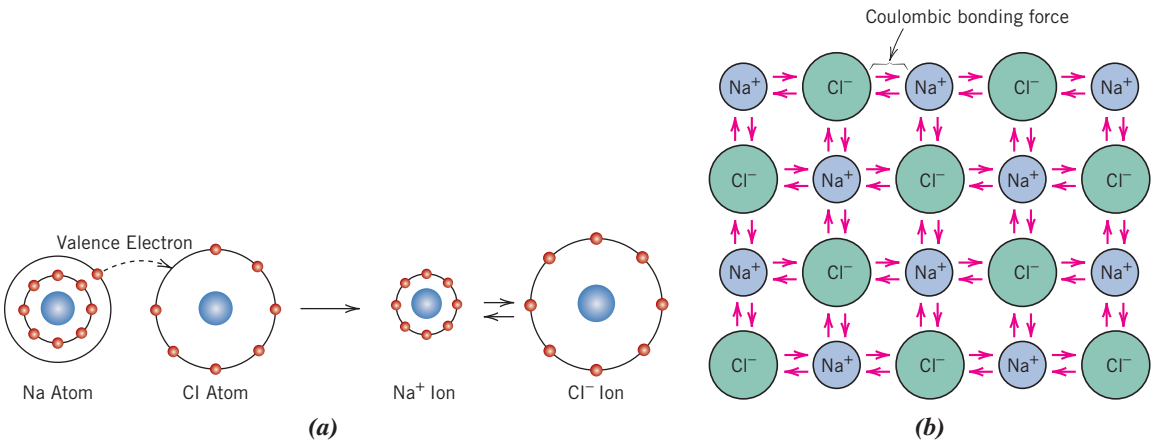


Figure 2.11 Schematic representations of (*a*) the formation of Na^+ and Cl^- ions and (*b*) ionic bonding in sodium chloride (NaCl).

Here ε_0 is the permittivity of a vacuum (8.85×10^{-12} F/m), $|Z_1|$ and $|Z_2|$ are absolute values of the valences for the two ion types, and e is the electronic charge (1.602×10^{-19} C). The value of A in Equation 2.9 assumes the bond between ions 1 and 2 is totally ionic (see Equation 2.16). Inasmuch as bonds in most of these materials are not 100% ionic, the value of A is normally determined from experimental data rather than computed using Equation 2.10.

An analogous equation for the repulsive energy is[6]

Repulsive energy—interatomic separation relationship

$$E_R = \frac{B}{r^n} \tag{2.11}$$

In this expression, B and n are constants whose values depend on the particular ionic system. The value of n is approximately 8.

Ionic bonding is termed *nondirectional*—that is, the magnitude of the bond is equal in all directions around an ion. It follows that for ionic materials to be stable, all positive ions must have as nearest neighbors negatively charged ions in a three-dimensional scheme, and vice versa. Some of the ion arrangements for these materials are discussed in Chapter 12.

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What Is Ionic Bonding?

Bonding energies, which generally range between 600 and 1500 kJ/mol, are relatively large, as reflected in high melting temperatures.[7] Table 2.3 contains bonding energies and melting temperatures for several ionic materials. Interatomic bonding is typified by ceramic materials, which are characteristically hard and brittle and, furthermore, electrically and thermally insulative. As discussed in subsequent chapters, these properties are a direct consequence of electron configurations and/or the nature of the ionic bond.

[6]In Equation 2.11, the value of the constant B is also fit using experimental data.

[7]Sometimes bonding energies are expressed per atom or per ion. Under these circumstances, the electron volt (eV) is a conveniently small unit of energy. It is, by definition, the energy imparted to an electron as it falls through an electric potential of one volt. The joule equivalent of the electron volt is as follows: 1.602×10^{-19} J = 1 eV.

Table 2.3
Bonding Energies and Melting Temperatures for Various Substances

Substance	*Bonding Energy (kJ/mol)*	*Melting Temperature (°C)*
	Ionic	
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF_2	1548	1418
	Covalent	
Cl_2	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830
	Metallic	
Hg	62	−39
Al	330	660
Ag	285	962
W	850	3414
	van der Waals[a]	
Ar	7.7	−189 (@ 69 kPa)
Kr	11.7	−158 (@ 73.2 kPa)
CH_4	18	−182
Cl_2	31	−101
	Hydrogen[a]	
HF	29	−83
NH_3	35	−78
H_2O	51	0

[a]Values for van der Waals and hydrogen bonds are energies *between* molecules or atoms (*inter*molecular), not between atoms within a molecule (*intra*molecular).

EXAMPLE PROBLEM 2.2

Computation of Attractive and Repulsive Forces between Two Ions

The atomic radii of K^+ and Br^- ions are 0.138 and 0.196 nm, respectively.

(a) Using Equations 2.9 and 2.10, calculate the force of attraction between these two ions at their equilibrium interionic separation (i.e., when the ions just touch one another).
(b) What is the force of repulsion at this same separation distance?

Solution

(a) From Equation 2.5b, the force of attraction between two ions is

$$F_A = \frac{dE_A}{dr}$$

Whereas, according to Equation 2.9,

$$E_A = -\frac{A}{r}$$

Now, taking the derivation of E_A with respect to r yields the following expression for the force of attraction F_A:

$$F_A = \frac{dE_A}{dr} = \frac{d\left(-\frac{A}{r}\right)}{dr} = -\left(\frac{-A}{r^2}\right) = \frac{A}{r^2} \tag{2.12}$$

Now substitution into this equation the expression for A (Equation 2.10) gives

$$F_A = \frac{1}{4\pi\varepsilon_0 r^2}(|Z_1|e)(|Z_2|e) \tag{2.13}$$

Incorporation into this equation values for e and ε_0 leads to

$$\begin{aligned} F_A &= \frac{1}{4\pi(8.85 \times 10^{-12}\ \text{F/m})(r^2)}[|Z_1|(1.602 \times 10^{-19}\ \text{C})][|Z_2|(1.602 \times 10^{-19}\ \text{C})] \\ &= \frac{(2.31 \times 10^{-28}\ \text{N}\cdot\text{m}^2)(|Z_1|)(|Z_2|)}{r^2} \end{aligned} \tag{2.14}$$

For this problem, r is taken as the interionic separation r_0 for KBr, which is equal to the sum of the K^+ and Br^- ionic radii inasmuch as the ions touch one another—that is,

$$\begin{aligned} r_0 &= r_{K^+} + r_{Br^-} \\ &= 0.138\ \text{nm} + 0.196\ \text{nm} \\ &= 0.334\ \text{nm} \\ &= 0.334 \times 10^{-9}\ \text{m} \end{aligned} \tag{2.15}$$

When we substitute this value for r into Equation 2.14, and taking ion 1 to be K^+ and ion 2 as Br^- (i.e., $Z_1 = +1$ and $Z_2 = -1$), then the force of attraction is equal to

$$F_A = \frac{(2.31 \times 10^{-28}\ \text{N}\cdot\text{m}^2)(|+1|)(|-1|)}{(0.334 \times 10^{-9}\ \text{m})^2} = 2.07 \times 10^{-9}\ \text{N}$$

(b) At the equilibrium separation distance the sum of attractive and repulsive forces is zero according to Equation 2.4. This means that

$$F_R = -F_A = -(2.07 \times 10^{-9}\ \text{N}) = -2.07 \times 10^{-9}\ \text{N}$$

Covalent Bonding

covalent bonding

A second bonding type, **covalent bonding,** is found in materials whose atoms have small differences in electronegativity—that is, that lie near one another in the periodic table. For these materials, stable electron configurations are assumed by the sharing of electrons between adjacent atoms. Two covalently bonded atoms will each contribute at least one electron to the bond, and the shared electrons may be considered to belong to both atoms. Covalent bonding is schematically illustrated in Figure 2.12 for a molecule of hydrogen (H_2). The hydrogen atom has a single 1*s* electron. Each of the atoms can acquire a helium electron configuration (two 1*s* valence electrons) when they share their single electron (right side of Figure 2.12). Furthermore, there is an overlapping of electron orbitals in the region between the two bonding atoms. In addition, the covalent

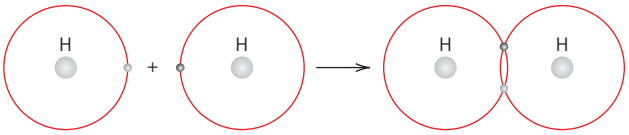


Figure 2.12 Schematic representation of covalent bonding in a molecule of hydrogen (H_2).

bond is *directional*—that is, it is between specific atoms and may exist only in the direction between one atom and another that participates in the electron sharing.

Many nonmetallic elemental molecules (e.g., Cl_2, F_2), as well as molecules containing dissimilar atoms, such as CH_4, H_2O, HNO_3, and HF, are covalently bonded.[8] Furthermore, this type of bonding is found in elemental solids such as diamond (carbon), silicon, and germanium and other solid compounds composed of elements that are located on the right side of the periodic table, such as gallium arsenide (GaAs), indium antimonide (InSb), and silicon carbide (SiC).

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What Is Covalent Bonding?

Covalent bonds may be very strong, as in diamond, which is very hard and has a very high melting temperature, >3550°C (6400°F), or they may be very weak, as with bismuth, which melts at about 270°C (518°F). Bonding energies and melting temperatures for a few covalently bonded materials are presented in Table 2.3. Inasmuch as electrons participating in covalent bonds are tightly bound to the bonding atoms, most covalently bonded materials are electrical insulators, or, in some cases, semiconductors. Mechanical behaviors of these materials vary widely: some are relatively strong, others are weak; some fail in a brittle manner, whereas others experience significant amounts of deformation before failure. It is difficult to predict the mechanical properties of covalently bonded materials on the basis of their bonding characteristics.

Bond Hybridization in Carbon

Often associated with the covalent bonding of carbon (as well other nonmetallic substances) is the phenomenon of *hybridization*—the mixing (or combining) of two or more atomic orbitals with the result that more orbital overlap during bonding results. For example, consider the electron configuration of carbon: $1s^2 2s^2 2p^2$. Under some circumstances, one of the 2*s* orbitals is promoted to the empty 2*p* orbital (Figure 2.13*a*), which gives rise to a $1s^2 2s^1 2p^3$ configuration (Figure 2.13*b*). Furthermore, the 2*s* and 2*p* orbitals can mix to produce four sp^3 orbitals that are equivalent to one another, have parallel spins, and are capable of covalently bonding with other atoms. This orbital mixing is termed *hybridization*, which leads to the electron configuration shown in Figure 2.13*c*; here, each sp^3 orbital contains one electron, and, therefore, is half-filled.

Bonding hybrid orbitals are directional in nature—that is, each extends to and overlaps the orbital of an adjacent bonding atom. Furthermore, for carbon, each of its four sp^3 hybrid orbitals is directed symmetrically from a carbon atom to the vertex of a tetrahedron—a configuration represented schematically in Figure 2.14; the angle between each set of adjacent bonds is 109.5°.[9] The bonding of sp^3 hybrid orbitals to the 1*s* orbitals of four hydrogen atoms, as in a molecule of methane (CH_4), is presented in Figure 2.15.

For diamond, its carbon atoms are bonded to one another with sp^3 covalent hybrids—each atom is bonded to four other carbon atoms. The crystal structure for diamond is shown in Figure 12.16. Diamond's carbon–carbon bonds are extremely strong, which accounts for its high melting temperature and ultrahigh hardness (it is the hardest of all materials). Many polymeric materials are composed of long chains of carbon atoms that are also bonded together using sp^3 tetrahedral bonds; these chains form a zigzag structure (Figure 14.1*b*) because of this 109.5° interbonding angle.

[8]For these substances, the *intra*molecular bonds (bonds between atoms in a molecule) are covalent. As noted in the next section, other types of bonds can operate between molecules, which are termed *inter*molecular.

[9]Bonding of this type (to four other atoms) is sometimes termed *tetrahedral bonding*.

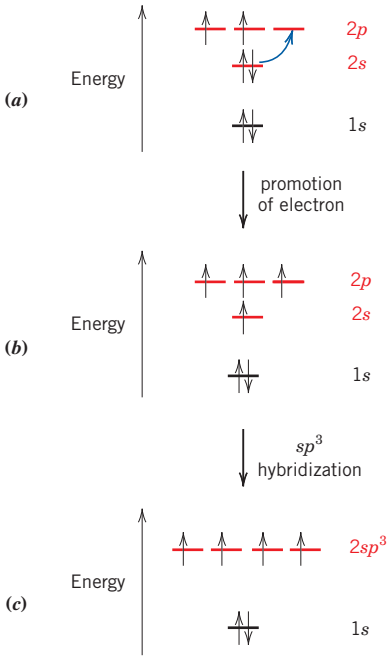


Figure 2.13 Schematic diagram that shows the formation of sp^3 hybrid orbitals in carbon. (*a*) Promotion of a 2*s* electron to a 2*p* state; (*b*) this promoted electron in a 2*p* state; (*c*) four $2sp^3$ orbitals that form by mixing the single 2*s* orbital with the three 2*p* orbitals.

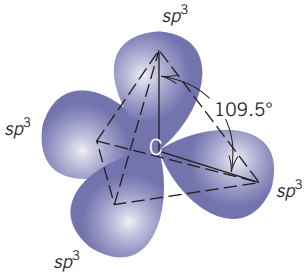


Figure 2.14 Schematic diagram showing four sp^3 hybrid orbitals that point to the corners of a tetrahedron; the angle between orbitals is 109.5°.
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.)

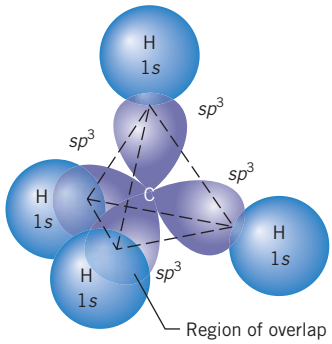


Figure 2.15 Schematic diagram that shows bonding of carbon sp^3 hybrid orbitals to the 1*s* orbitals of four hydrogen atoms in a molecule of methane (CH_4).
(From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes,* 4th edition, 2004. Reprinted with permission of John Wiley & Sons, Inc.)

Other types of hybrid bonds are possible for carbon, as well as other substances. One of these is sp^2, in which an *s* orbital and two *p* orbitals are hybridized. To achieve this configuration, one 2*s* orbital mixes with two of the three 2*p* orbitals—the third *p* orbital remains unhybridized; this is shown in Figure 2.16. Here, $2p_z$ denotes the unhybridized *p* orbital.[10] Three sp^2 hybrids belong to each carbon atom, which lie in the same plane such that the angle between adjacent orbitals is 120° (Figure 2.17); lines drawn from one orbital to another form a triangle. Furthermore, the unhybridized $2p_z$ orbital is oriented perpendicular to the plane containing the sp^2 hybrids.

These sp^2 bonds are found in graphite, another form of carbon, which has a structure and properties distinctly different from those of diamond (as discussed in Section 12.4). Graphite is composed of parallel layers of interconnecting hexagons. Hexagons form from planar sp^2 triangles that bond to one another in the manner presented in Figure 2.18—a carbon atom is located at each vertex. In-plane sp^2 bonds are strong; by way of contrast, weak interplanar bonding results from van der Waals forces that involve electrons originating from the unhybridized $2p_z$ orbitals. The structure of graphite is shown in Figure 12.17.

[10]This $2p_z$ orbital has the shape and orientation of the p_z shown in Figure 2.5*c*. In addition, the two *p* orbitals found in the sp^2 hybrid correspond to the p_x and p_y orbitals of this same figure. Furthermore, p_x, p_y, and p_z are the three orbitals of the sp^3 hybrid.

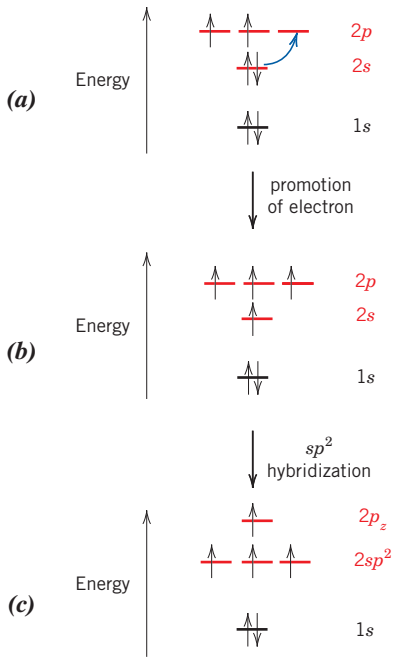


Figure 2.16 Schematic diagram that shows the formation of sp^2 hybrid orbitals in carbon. (*a*) Promotion of a 2*s* electron to a 2*p* state; (*b*) this promoted electron in a 2*p* state; (*c*) three $2sp^2$ orbitals that form by mixing the single 2*s* orbital with two 2*p* orbitals—the $2p_z$ orbital remains unhybridized.

Metallic Bonding

metallic bonding

Metallic bonding, the final primary bonding type, is found in metals and their alloys. A relatively simple model has been proposed that very nearly approximates the bonding scheme. With this model, these valence electrons are not bound to any particular atom in the solid and are more or less free to drift throughout the entire metal. They may be thought of as belonging to the metal as a whole, or forming a "sea of electrons" or an "electron cloud." The remaining nonvalence electrons and atomic nuclei form what are called *ion cores*, which possess a net positive charge equal in magnitude to the total valence electron charge

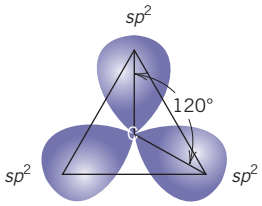


Figure 2.17 Schematic diagram showing three sp^2 orbitals that are coplanar and point to the corners of a triangle; the angle between adjacent orbitals is 120°.
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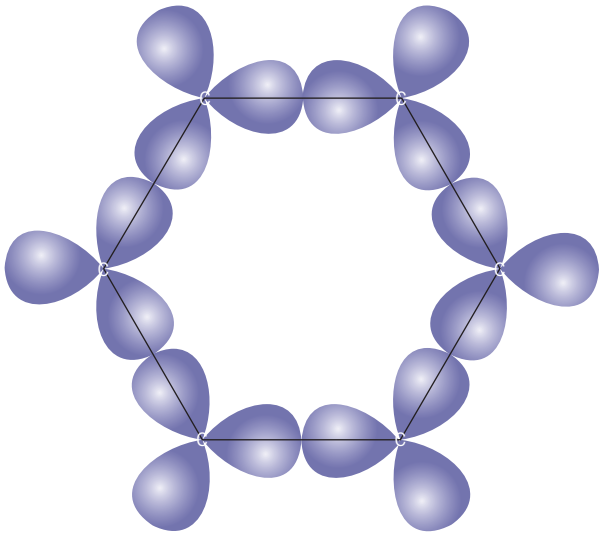


Figure 2.18 The formation of a hexagon by the bonding of six sp^2 triangles to one another.